INCH-POUND

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MILITARY SPECIFICATION

LOGIC ASSEMBLY
(For Exercise Head Mk 85)

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for procurement of two types of Logic Assembly (see 6.1.1).

1.2 Classification. The Logic Assembly shall be classified as follows:

TYPE I Logic Assembly PL 5547531

TYPE II Logic Assembly PL 6098364

2. APPLICABLE DOCUMENTS

2.1 Government documents,

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization Branch (3730), Indian Head, Maryland 20640-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1355

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SPECIFICATIONS

MILITARY

MIL-P-116 Preservation, Methods of

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (Attn: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified (see 6.2), the issues are those cited in the solicitation.

DRAWINGS

NAVAL ORDNANCE SYSTEMS COMMAND (CAGE Code 10001)

DL 1330422	Logic Assembly shipping (packing) condition
2540510	Exercise Head Test Set
2540522	Test Cable
PL 5547531	Logic Assembly
PL 6098364	Logic Assembly

(Application for copies of NAVSEA drawings should be addressed to: Commanding Officer, Naval Ordnance Station, (ATTN: Code 802), Louisville, KY 40214-5001.)

2.2 Order of precedence, In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.4) in accordance with 4.5.

3.2 Material and treatments of materials and parts. Unless otherwise specified in the contract or order, all materials and treatments of materials and parts used in the manufacture of the Logic Assembly shall be in accordance with requirements specified herein and with the applicable documents listed in PL 5547531 or PL 6098364.

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3.3 Design and construction. The design and construction of the Logic Assembly, Type I and Type II, shall be in accordance with PL 5547531 or PL 6098364 (see 6.2).

3.4 Performance characteristics.

NOTE: The numerical values tabulated in column I of the following tables are applicable to the tests conducted in accordance with the acceptance test and examinations of 4.7. The values in column II are applicable to the tests of paragraph 4.5.2.3.2. and table VIII, test sequence 10. Minimum and maximum values apply to absolute values only (i.e., -40.6 volts direct current (Vdc) is greater than -40.1 Vdc).

3.4.1 Inputs. The Logic Assembly shall require or accept the inputs given in table I and shall meet all functional requirements with the dc inputs given:

TABLE I. DC inputs.

Input		Column I	Column II
Voltage	+40 Vdc	+40 ± 2 Vdc	+40 ± 2 Vdc
	-40 Vdc	-40 ±2 Vdc	-40 ±2 Vdc
Battery	+5.4V (center tapped)	+5.4 Vdc nominal +4.4 Vdc minimum	+5.4 Vdc nominal +4.0 Vdc minimum
Current	+40 Vdc		
	-40 Vdc		

3.4.1.1 Signal inputs. The Logic Assembly shall operate with the signal inputs given in table II. Those signals that are digital in their operation are shown in both states.

TABLE II. Signal inputs.

Input	Column I	Column II
32 Vac	32 ± 4 Vrms 250 mw 1900 ± 200 Hz	32 ± 4 Vrms, 250 mw 1900 ± 200 Hz
4.8 kHz in phase	26.0 ± 1.0 Vrms 4800 ± 10 Hz or 2000 ± 20 Hz	26.0 ± 2.0 Vrms 4800 ± 200 Hz or 2000 ± 20 Hz

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TABLE II. Signal inputs - Continued.

Input	Column I	Column II
Inst course gyro	0.0 to 3.5 Vrms at 4.8 kHz or 2.0 kHz	0.0 to 3.5 Vrms at 4.8 kHz or 2.0 kHz
R > B warshot load	-11 Vdc min	-10 Vdc min
Electronic drive out	5 Vrms, 2 kHz	5 Vrms, 2 kHz
DCO switch	-5.4 Vdc nominal	-5.4 Vdc nominal
ALD/HLD reset	-5.4 Vdc nominal	-5.4 Vdc nominal
Steeraway disable 200 ft switch	-5.4 Vdc nominal 0.0 ± 0.2 Vdc max open	-5.4 Vdc 0.0 ± 0.2 Vdc max open

3.4.2 Outputs. The signal outputs given in table III shall be provided by the Logic Assembly.

TABLE III. Signal outputs.

Output	Column I	Column II
ALD/HLD	0.0 ± 0.2 Vdc max open	0.0 ± 0.2 Vdc max open
-10 Vdc	-10.0 ± 0.5 Vdc	-10.0 ± 0.5 Vdc
Course gyro demod out	See 3.4.3.1	See 3.4.3.1
RPM (MK 46 MOD 1)	See 3.4.3.2	See 3.4.3.2
Electronic drive in	5 Vrms, 2 kHz (in or out of phase with input)	5 Vrms, 2 kHz (in or out of phase with input)
Time share trigger	+5.5 + 1.3, -1.0 volt pulse (2 ms duration) 0.50 ± 0.01 sec rep rate	+5.5 + 1.3, -1.0 volt pulse (2 ms duration) 0.50 ± 0.01 sec rep rate

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TABLE III. Signal outputs - Continued.

Output	Column I	Column II
Inst timing mark	+5.0 ± 1.5 volt pulse (4 ms duration) 0.50 ± 0.01 sec rep rate	+5.0 ± 1.5 volt pulse (4 ms duration) 0.50 ± 0.01 sec rep rate
Power failure squib	-40 ± 2 Vdc exponential pulse	-40 ± 2 Vdc exponential pulse
DCO squib	1 amp min into 1 ohm	1 amp min into 1 ohm
Term squib	1 amp min into 1 ohm	1 amp min into 1 ohm

3.4.3 Performance characteristics. The performance characteristics presented in tables IV and V shall occur after the Logic Assembly has been activated. The logic assembly utilizes the inputs in tables IV and V to produce the outputs listed in columns I and II.

TABLE IV. Course gyro demod out performance.

Inputs	Column I	Column II
a) 0.0 Vrms, 4.8 kHz	0.0 ± 0.010 Vdc	0.0 ± 0.02 Vdc
b) 5.0 Vrms, 4.8 kHz	-0.379 ± 0.034 Vdc	-0.379 ± 0.068 Vdc
c) 10.0 Vrms, 4.8 kHz	-0.750 ± 0.079 Vdc	-0.750 ± 0.158 Vdc

TABLE V. RPM (Mk 46 Mod 1) performance.

Inputs	Column I	Column II
a) 32 ± 4 Vrms, 2000 Hz	0.0 ± 0.1 Vdc	0.0 ± 0.2 Vdc
b) 32 ± 4 Vrms, 1900 Hz	-0.45 ± 0.10 Vdc	-0.45 ± 0.2 Vdc
c) 32 ± 4 Vrms, 1800 Hz	-0.85 ± 0.15 Vdc	-0.9 ± 0.2 Vdc

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3.4.4 Life. Life requirements shall be as follows:

3.4.4.1 Operating. The Logic Assembly shall meet requirements of 3.4.1 through 3.4.3 operation throughout its mission time of 9 minutes.

3.4.4.2 Total operating life. The Logic Assembly shall perform satisfactory operation for a minimum of 200 cycles of operation as defined herein and for a period of 200 hours continuous operation with the application of steady state power. A cycle of operation shall be 9 minutes with a cooling off period of 10 minutes preceding each cycle. The life requirement shall begin at the date of acceptance by the contracting activity. Individual items with lesser life expectancy may be replaced or disabled upon conclusion of their respective life.

3.5 Environmental requirements. Unless otherwise specified, the Logic Assembly shall conform to performance and product characteristics requirements of this specification after being exposed, at any temperature in the applicable range, to the environments under all conditions specified in this section.

3.5.1 Vibration. The Logic Assembly shall withstand the vibration shown in table VI.

TABLE VI. Vibration schedule.

Condition (See 6.5)	Double Amplitude (in) minimum	Acceleration (g's) minimum	Frequency (Hz)
Nonoperating	--	1.0	5 to 60
Ready nonoperating	0.2	--	5 to 17
Ready	--	3.0	17 to 500

3.5.2 Shock. The Logic Assembly, packaged in accordance with DL 1330422 (see 5.1.1) shall withstand the shock of drop on a nonyield surface from a height of 36 inches. Additionally, the Logic Assembly, in the condition shown in table II, shall withstand shocks specified in table VII.

3.5.3 Temperature. The Logic Assembly shall withstand the specified temperatures in the following specified conditions.

3.5.3.1 Nonoperating and ready nonoperating. The Logic Assembly shall withstand ambient temperatures between -45 degrees Fahrenheit (°F) and 140°F wherein the duration of the upper temperature does not exceed 4 hours at any one time, and the temperature is stabilized in a temperature range of 60 to 95°F prior to any performance test.

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TABLE VII. Shock schedule.

Condition (See 6.5)	Peak Magnitude (g's)	Duration (milliseconds)	Acceleration Direction 1/	Number of Impacts
Ready nonoperating	60 ± 10%	8.0 to 10.0	3 fwd of aft axis 3 left or right axis 3 up or down axis	9
Ready	45 ± 10%	18.0 to 20.0	Long. fwd axis	3
	85 ± 10%	0.5 to 1.5	3 up or down axis 3 right or left axis	6
	300 ± 10%	0.5 to 1.5	Long. aft axis	3

1/ See figure 1 for axes of Logic Assembly.

3.5.3.2 Ready and operating. The Logic Assembly, when operated in an environment with a temperature between 32°F and 98°F, shall meet the performance and product characteristics of this specification (column II values) after exposure to temperatures between 0°F and 140°F, for a period wherein the durations of the upper temperature limit does not exceed 4 hours at any one time.

3.5.4 Pressure. The Logic Assembly shall withstand external pressures of from 2.7 to 20.0 pounds per square inch (lb/in^2) absolute.

3.5.5 Humidity. The Logic Assembly shall withstand exposure to 95 percent relative humidity at 110°F for 8 hours without degradation of finish or functional capabilities,

3.6 Workmanship. workmanship and finish shall be of sufficiently high grade to assure satisfactory operation consistent with requirements of this specification. All parts used in the logic assembly shall be thoroughly clean and free of excess materials, chips, and loose or spattered foreign materials. Standards of workmanship exhibited in any approved first article sample, subject to any qualifications stated in the Government's notice of approval, shall be determinative of requirements of the contract relative to workmanship insofar as not specifically covered by applicable specifications.

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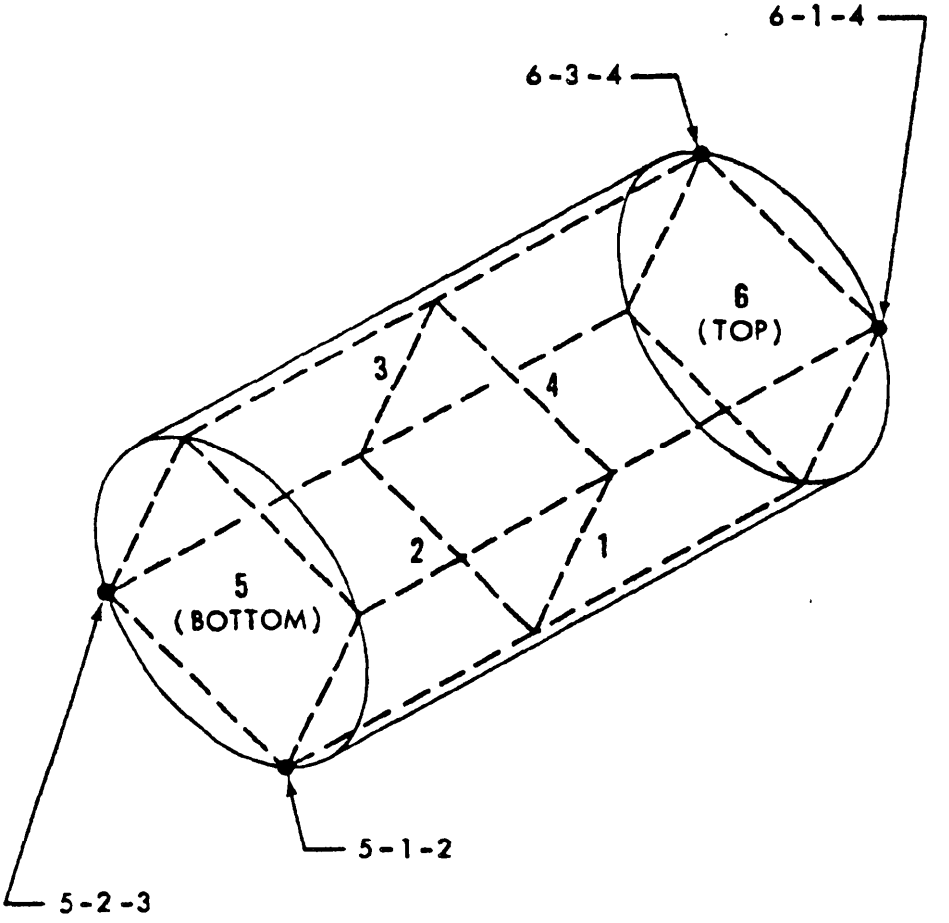


FIGURE 1. Definition of faces and corners of shipping container DL 1330422.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection, Unless otherwise specified in the contract or purchase order (see 6.2), the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products and supplies submitted to the Government for acceptance comply with all requirements of the contract, Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements; however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of Inspections. Inspections of the Logic Assembly shall be classified as follows:

a. First article inspection (see 4.5).

b. Periodic production inspection (see 4.6).

c. Quality conformance inspection (see 4.7).

4.3 Test conditions. Unless otherwise specified, the Logic Assembly shall be subjected to tests under following conditions:

a.	Temperature	Room ambient (60 to 95°F)
b.	Humidity (relative)	Room ambient, up to 95 percent
c.	Altitude	Normal ground atmospheric pressure
d.	Vibrations	None

4.4 Test reports. When specified in the contract or purchase order (see 6.2), a test report shall be prepared (see. 6.3) covering the results of tests specified in 4.5, first article. A second test report shall be prepared covering 4.6 and 4.7 quality conformance tests and inspection, and forwarded to the contracting activity (see 6.3).

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4.5 First article sample. A first article sample of three Logic Assemblies, manufactured in accordance with 3.3, shall be subjected to first article tests specified in table VIII, in the listed sequence, at an activity designated by the contracting activity. Acceptance of the first article sample shall be based on no defects in the sample. Further production of the Logic Assembly by the contractor, prior to approval of the first article sample, shall be at contractor's risk.

TABLE VIII. First article test schedule.

Sequence	Test Description	Logic Assembly Number		
		1	2	3
1	Acceptance Tests and Examinations (4.7)	x	x	x
2	Vibration (4.5.2.1)	x		x
3	Shock (4.5.2.2)	x	x	
4	Temperature, storage (4.5.2.3.1)	x	x	
5	Temperature, operating (4.5.2.3.2)	x	x	
6	Pressure test (4.5.2.4)	x	x	
7	Humidity test (4.5.2.5)	x		x
8	Life test (4.5.2.6)	x		
9	Acceptance Tests and Examinations (4.7)		x	x
10	Acceptance Tests and Examinations (4.7) with column II values specified in 3.4	x		

4.5.1 First article test equipment. The following items of test equipment are required to perform first article tests specified herein:

a. Vibration table that can vibrate the Logic Assembly, plus fixture weight at 3 gravitational acceleration units (g's) maximum over a frequency range of 5 to 500 Hz with a maximum double amplitude of 0.2 inch.

b. A shock testing machine that can apply shocks of up to 300 g's maximum at durations from 0.5 to 20 milliseconds.

c. A temperature chamber that can program temperature variations over a range of -45 to +140°F.

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d. A humidity chamber that can provide a constant humidity condition of 95 percent at 110°F for extended periods up to 8 hours.

4.5.2 Description of tests.

4.5.2.1 Vibration,

4.5.2.1.1 Nonoperating. The packaged Logic Assembly (see 5.1.1) shall be placed on a vibration platform that is fenced on all sides so that the pack cannot tip over or fall off the platform. With the packaged Logic Assembly in the vertical position, gradually increase the frequency from 5 Hz until the package just begins to leave the platform. At this point, run test for 1 hour. Repeat procedure with the packaged Logic Assembly laying on its side.

4.5.2.1.2 Ready nonoperating and ready. The unpackaged Logic Assembly mounted on a support frame, shall be rigidly attached to the vibration equipment. This support frame shall transmit the vibrations listed in table VI to the Logic Assembly along the axes shown in figure I. Vibrations of table VI shall be accomplished by cycling, the frequency rate being logarithmic with 15 minutes required to proceed from the lower to higher and back to the lower extremity of the frequency range. The total vibration time in each axis shall be 180 minutes (12 cycles), in a room ambient condition.

4.5.2.2 Shock.

4.5.2.2.1 Nonoperating. The Logic Assembly in level A packaging (see 5.1.1) shall be subjected to free fall drop tests on corners and surfaces of the container. The packaged unit shall be dropped from a height of 36 inches above an impact surface which shall be essentially nonyielding stone, concrete or similar material. The container, with packaged unit inside, shall be dropped in sequence on corners 5-1-2, 6-3-4, 5-2-3 and 6-1-4, and side 1 and face 5 as defined in figure 2.

4.5.2.2.2 Ready nonoperating and ready. Mount the unpackaged Logic Assembly onto a shock test fixture so that a minimum of 90 percent of table input of the fixture will be applied to the Logic Assembly under test. Subject the Logic Assembly to shocks listed in table VII, in the conditions specified. Duration shall be measured by an accelerometer mounted on input table of fixture to indicate applied shock.

4.5.2.3 Temperature.

4.5.2.3.1 Temperature, nonoperating. The Logic Assembly preserved and packaged in accordance with 5.1.1, shall be placed in a temperature controlled chamber. The chamber shall then be subjected to the following temperature cycles:

a. Heat chamber to +140 +5, -0°F in 2.5 ± 0.5 hours.

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b. Maintain at +140 +5, -0°F for 4 hours minimum, then within 2.5 ± 0.5 hours cool chamber to -45 +1, -5°F.

c. Maintain at -45 +0, -5°F for 4 hours minimum, then within 2.5 ± 0.5 hours return chamber to room temperature.

Repeat this schedule for a minimum of three cycles.

4.5.2.3.2 Temperature, operating. The Logic Assembly, mounted as described in figure 1, shall meet the quality conformance test requirements of 4.7 except as specified below.

a. The Logic Assembly shall be stabilized for 1 hour at a temperature of 140 +5, -0°F, then transferred to an environment at 98 +5, -0°F and tested to assure conformance with performance and product character-istics of this specification (column II values). This conformance shall be shown to exist for any combination of power supply and input voltages in table I (Column II) and table II (column II) of this specification.

b. The Logic Assembly shall be stabilized for 1 hour at a temperature of O +0, -5°F, then transferred to an environment at 32 +0, -3°F and tested to assure conformance with performance and product character-istics of this specification (column II values) . This conformance shall be shown to exist for any combination of power supply and input voltages in table I (column II) and table II (column II) of this specification.

4.5.2.4 Pressure. Subject the unpackaged Logic Assembly to absolute pressures of 2.7 lb/in^2 or less and 20.0 lb/in^2 or more, sustained for 30 minutes at each pressure level.

4.5.2.5 Humidity. The unpackaged Logic Assembly shall be placed in a suitable test chamber and subjected to at least 95 percent relative humidity at a temperature of 110 +5, -0°F for a period of 8 hours minimum. Upon completion of this test, remove the Logic Assembly from the chamber and immediately initiate the acceptance test (see 4.7).

4.5.2.6 Life test.

4.5.2.6,1 Cycling test. The Logic Assembly shall be energized for 200 cycles wherein the duty cycle shall be 9 minutes on and 10 minutes off. On each cycle all functions shall be properly loaded and energized.

4.5.2.6.2 Steady state test. The Logic Assembly shall be operated for a period of 200 continuous hours with steady state power applied,

4,6 Periodic production. Periodic production samples shall be selected from units having satisfactorily passed the quality conformance tests and examinations of 4.7. These samples shall be subjected to the tests of table

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IX, in the order specified, at an activity designated by the contracting activity. Periodic production sampling shall be as follows:

a. Step 1. Upon initiation of production, randomly select 1 unit from each group of 30 units. Upon satisfactory compliance of samples selected from 3 consecutive groups, shift to Step 2. Each sample so selected shall be numbered consecutively starting with Sample Number 1.

b. Step 2. Randomly select 1 unit from each group of 70 units. Continue with Step 2 for the balance of the contract or until a defective unit is found. Each sample so selected shall be numbered consecutively starting with a sample number one greater than the highest sample number selected in Step 1.

TABLE IX. Periodic production test schedule.

Test	Sample and Test Sequence		
	Sample No. 1,4, Etc.	Sample No. 2,5 Etc.	Sample No. 3,6, Etc.
Nonoperating vibration (4.5.2.1)	1[1/]		1[1/]
Shock (4.5.2.2)		1[1/]	3[1/]
Temperature nonoperating (4.5.2.3.1)		2	4
Hi-Temperature operating (4.5.2.3.2, a)		3	5
Lo-Temperature operating (4.5.2.3.2, b)	3		2
Pressure (4.5.2.4)	2		6
Humidity (4.5.2.5) Life (4.5.2.6)			7 9
Acceptance (Column I) (4.7)			8
Acceptance (Column II) (4.7)			10

[1/] Visual examination required after test.

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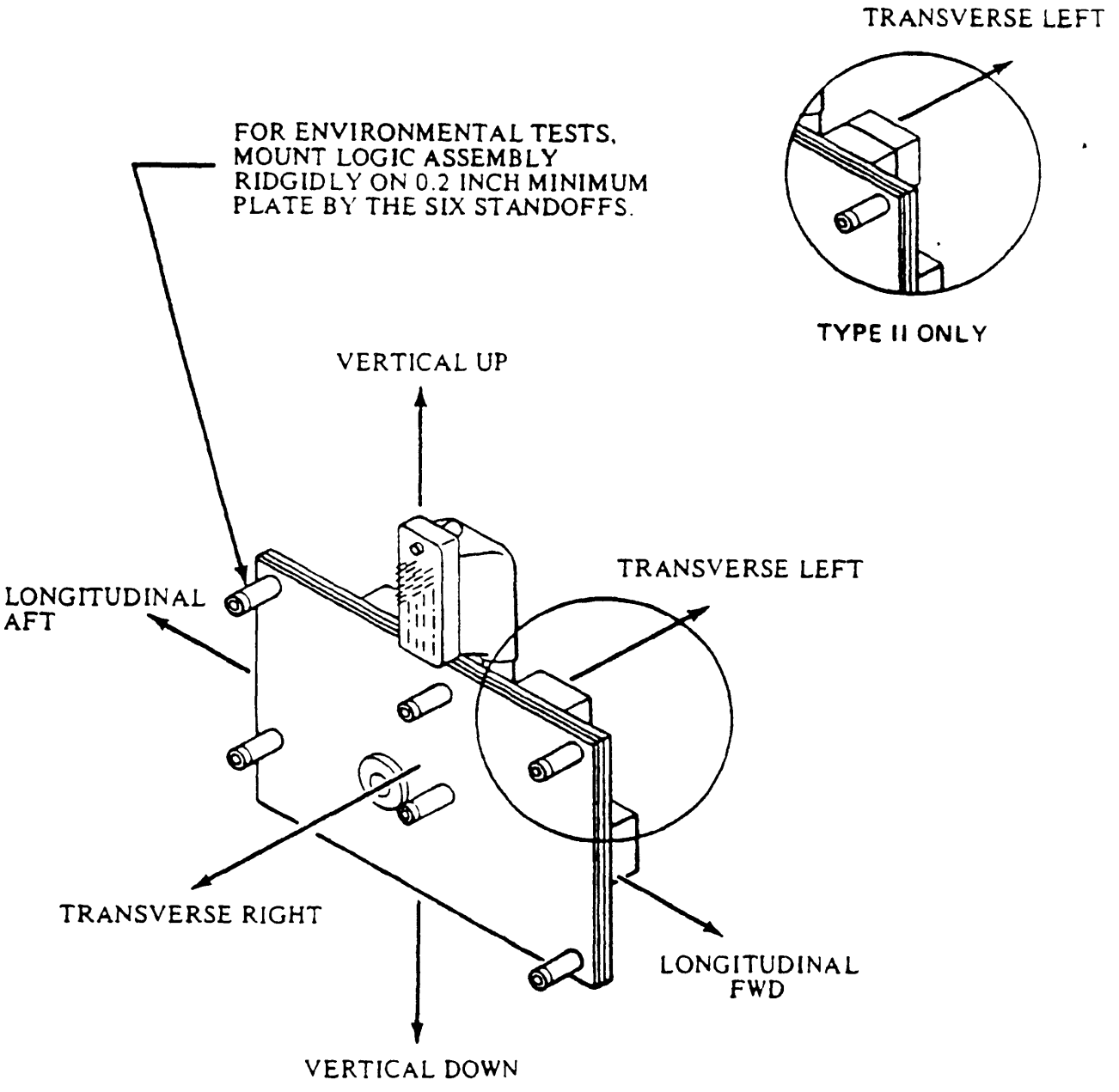


FIGURE 2. Definition of faces and corners of shipping container DL 1330422.

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4.6.1 Defective units. When a defective unit (see 6.5.11) is found in any group, acceptance of all units on hand shall be withheld and the contractor shall:

a. Determine the cause of the defect.

b. Take the necessary actions to correct or eliminate the defect from all units on hand and, if so directed by the contracting activity, from those units previously shipped.

c. Submit to the contracting activity for approval, the proposed corrective action to prevent recurrence of the defect.

d. Submit to the contracting activity for approval the proposed retest program to verify that the corrective action eliminated the defect.

e. Return to Step 1 and proceed as indicated when inspection is resumed.

4.7 Quality conformance. Tests and examinations specified in 4.7.3 shall be conducted on each Logic Assembly to be delivered. Unless otherwise specified (see 6.2), the sequence of testing must be the order shown herein. Defective units shall be rejected.

4.7.1 Quality conformance criteria. Failure of the end product to pass any examination or test of this section will automatically classify the unit as nonconforming. When nonconforming units are corrected by the contractor, the unit shall be reinspected. The reinspection of a nonconforming unit may be limited to the test or examination which defined the nonconformance, or when so directed by the cognizant inspector, a complete re-examination and retest of the unit may be required. Nonconforming units, which have not been corrected, will be considered for acceptance only upon formal application by the contractor to the contracting activity.

4.7.2 Quality conformance test equipment. Items of test equipment shown in table X are required to perform acceptance tests specified herein. All items of test equipment shall be of sufficient accuracy and precision to permit valid performance of required tests. Test instruments shall be calibra-ted at regular intervals. Procedures for and results of all calibrations shall be subject to approval of the cognizant Government inspector. Items having equivalent characteristics may be substituted where necessary. Verify that all test equipment used is identified as being within calibration.

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TABLE X. Quality conformance test equipment.

Item	Identification
Exercise Head Test Set	Drawing 2540510
Test Cable	Drawing 2540522
Electronic Counter (Range: 10 to 120 kHz Accuracy: ± 1 count)	Hewlett-Packard, Model 522B or equal
Oscilloscope (Bandpass: dc to 15 megacycles Rise time: from 0.023 sec. Accuracy: 3 percent from 0.05 to 20 volts/cm.)	Tektronix, Model 535A (with type CA preamplifier) or equal
Digital DC Voltmeter (Range: 0.001 to 999 Vdc Accuracy: ± 0.2 percent of reading, ± 1 count).	Hewlett-Packard, Model 405BR or equal
AC vtvm (Range: 1.0 mV to 300 Vrms Accuracy: ± 1 percent 50 to 500 Hz. Frequency Range: 10 to 4 Hz.)	Hewlett-Packard, Model 400H or equal

4.7.3 Test procedure.

4.7.3.1 Initial test setup.

a. Position main power circuit breaker on test set to "off".

NOTE: Verify that exercise head is not connected to test set.

b. Connect Logic Assembly to exercise head test set drawing 2540510 as shown in figure 3.

c. Position controls on exercise head test set panel as listed in table XI.

d. Position main power circuit breaker on test set to “on” and connect digital dc voltmeter to meter monitor jacks on test set panel.

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e. Position dc volts switch to +40 and adjust 40V power supply for 40.0 ± 0.5 Vdc indication on digital dc voltmeter.

f. Position dc volts switch to -40 and adjust -40V power supply for -40.0 ± 0.5 Vdc indication on digital dc voltmeter.

TABLE XI. Exercise head test set control positions (initial setup).

Control	Position
Galvo switches	center
dc input calibrate control	500
ac input calibrate control	500
oscilloscope lamp switch	down
exercise head pwr switch	down
range switch	R>B
logic power switch	off
turnaway ko switch	up
dc volts switch	off
input sound level switch	-40
ac select switch	input sound
rpm freq switch	2000
rpm level control	32
oscilloscope motor switch	down
200 ft switch	down[1/]

1/ Type II only

4.7.3.2 Time mark generator circuit.

a. Position logic power switch to “on” and connect oscilloscope between “timing marks” and “ground” jacks on test set panel. Oscilloscope shall display 3.5 to 6.5 volt pulses.

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b. Remove oscilloscope from “timing marks” and “ground” jack and connect electronic counter to these jacks. Counter shall display a time measurement of 490 to 510 milliseconds,

c. Remove counter from “timing marks” and connect it between "tssf trig” and “ground” jacks on test set panel. Counter shall display a time measurement of 490 to 510 milliseconds.

d. Remove counter from "tssf" and “ground” jacks and connect oscilloscope between these jacks. Oscilloscope shall display 4.5 to 6.8 volt pulses.

4.7.3.3 Course gyro demodulator circuit.

a. Position ac select switch to "demod" and connect digital dc voltmeter between "demod" and “ground” jacks on test set panel. Connect ac vtvm between "ac input calibrate” and “ground” jacks on test set panel.

NOTE: AC INPUT CALIBRATE dial shall be set above 500 for 0°, 0 root-mean-square voltage and below 500 for 180°, 0 rms voltage.

b. Position course gyro switch to “down” and adjust ac input calibrate control for a null indication on ac vtvm. Digital dc voltmeter shall indicate -0.01 to +0.01 Vdc.

c. Adjust ac input calibrate control for 5.0 $\pm$ 0.1 Vrms (0° O) indication on ac vtvm. Digital dc voltmeter shall indicate -0.345 to -0.414 Vdc.

d. Adjust ac input calibrate control for 10.0. $\pm$ 0.1 Vrms (0° O) indication on ac vtvm. Digital dc voltmeter shall indicate -0.671 to -0.828 Vdc.

e. Return course gyro switch to center position.

4.7.3.4 RPM tachometer circuit.

a. Connect digital dc voltmeter between "rpm" and “ground” jack on test set panel. Position rpm switch to down and ac select switch to "rpm" .

b. Verify that rpm freq switch on test set panel is set at 2000, and rpm level control on test set panel is set at 32. Digital dc voltmeter shall indicate -0.1 to +0.1 Vdc.

c. Vary rpm level control from 28 to 36. Digital dc voltmeter shall not vary more than 10 mV.

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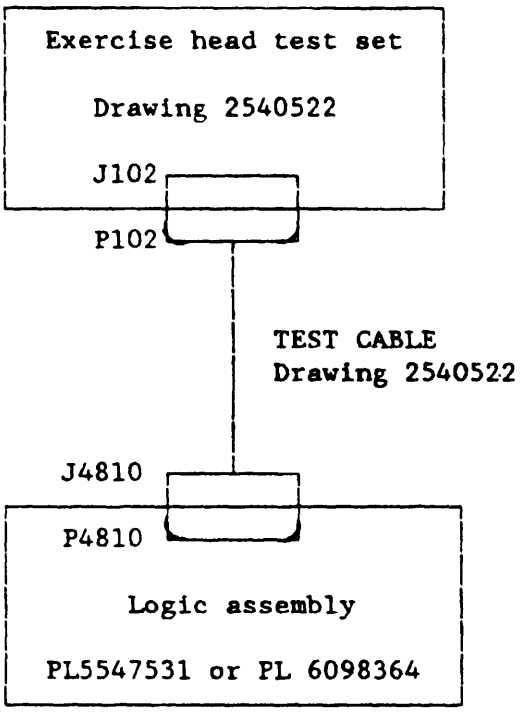


FIGURE 3. Logic Assembly test setup.

d. Position rpm level control to 32 and rpm freq switch to 1900. Digital dc voltmeter shall indicate -0.35 to -0.55 Vdc.

e. Vary rpm level control from 28 to 36. Digital dc voltmeter indication shall not vary more than 10 mv.

f. Position rpm level control to 32 and rpm freq switch to 1800. Digital dc voltmeter shall indicate -0.7 to -1.0 Vdc.

g. Vary rpm level control from 28 to 36. Digital dc voltmeter indication shall not vary more than 10 mv.

h. Return rpm switch to center position.

4.7.3.5 Turnaway circuit.

a. Verify that normal steering indicator is illuminated.

b. Position range switch to R<B. Normal steering indicator shall extinguish and reverse steering indicator shall illuminate.

c. Position turnaway ko switch to “down”. Reverse steering indicator shall extinguish and normal steering indicator shall illuminate.

d. Position turnaway ko switch to “up” and range switch to R>B.

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4.7.3.6 Squib logic circuit.

4.7.3.6.1 Connect digital dc voltmeter between “aid ind" jack and “ground” jack on test set panel.

4.7.3.6.2 Hold inertia switch down momentarily. DCO squib, termination squib, and recovery indicators shall illuminate.

4.7.3.7 Power failure circuit.

a. Position logic power switch to "off”. Power failure indicator shall illuminate for 4 to 6 seconds. Digital dc voltmeter shall indicate +39.0 to +41.0 Vdc.

b. Hold dco-ald reset switch momentarily to ald reset, Digital dc voltmeter shall indicate -0.01 to +0.01 Vdc.

TYPE II ONLY. Position 200 ft. switch to “on”. DCO squib indicator shall illuminate. Position logic power switch to “on”. DCO squib indicator shall extinguish. Position logic power switch to “off”. DCO squib indicator shall illuminate. Position 200 ft. switch to “off”. DCO squib indicator shall extinguish.

c. Position logic power switch to “on” and momentarily hold dco-ald reset switch to dco position. Dco squib and termination squib indicators shall illuminate when switch is in dco position, Position logic power switch to “off”.

d. Position logic power switch to “on”. After 8 to 9 seconds has elapsed, position logic power switch to “off”. Power failure indicator shall not illuminate.

e. Position logic power switch to “on”. After 11 to 12 seconds have elapsed, position logic power switch to “off”. Power failure indicator shall illuminate.

4.7.3.8 Minus 10 volt regulator circuit.

a. Connect digital dc voltmeter to meter monitor jack and position dc volts switch to -10.

b. Position logic power switch to “on”. Digital dc voltmeter shall indicate -9.5 to -10.5 Vdc.

c. Position dc volts switch to -40 and adjust -40V power supply on test set for -38 $\pm$ 0.5 Vdc indication on digital dc voltmeter. Position dc volts switch to -10 and record voltage indicted on digital dc voltmeter.

d. Position dc volts switch to -40 and adjust -40V power supply for -42.0 $\pm$ 0,5 Vdc indication on digital dc voltmeter, Position dc

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volts switch to -10 and record voltage indicated on digital dc voltmeter.

e. The difference between digital dc voltmeter indication in 4.7.3.8 (c and d) shall not exceed 0.1 Vdc.

f. Repeat procedures in 4.7.3.1 e and f.

g. Position all power switches to off and disconnect test equipment.

4.7.4 Visual inspections.

4.7.4.1 Identification marking. Nomenclature, control callout and identification data shall be correct, clear and legible in accordance with PL 5547531 or PL 6098364.

4.7.4.2 Interchangeability.

a. Connector P4810 shall accept mating unit without resorting to selection, modification, or fitting in accordance with PL 5547531 or PL 6098364.

b. Length shall be 5.99 inches maximum in accordance with PL 5547531 or PL 6098364.

c. Width shall be 3.99 inches maximum in accordance with PL 5547531 or PL 6098364.

4.7.4.3 Packaging. Packaging shall be inspected for conformance to the requirements of Section 5.

4.7.5 Process verification. Criteria stipulated in this paragraph and its subparagraphs are applicable to the production process instead of specific units of product. Inspections stipulated are general in scope and technique and frequency of verification will be contingent upon specific process conditions. Positive action by the contractor is required when process criteria findings indicate poor inspection.

4.7.5.1 Lacing. Lacing and cable configuration shall be in accordance with procedures approved by the contracting activity. Specific methods of verifying adequacy of cable lacing shall include visual and dimensional verification.

4,7.5.2 Soldering. Soldered joints shall be in accordance with good commercial practices. Unsoldered connections or those of a “cold” solder or “rosin” joint shall be considered defective.

4.7.5,3 Cleanliness. Presence of foreign material in assembly shall classify unit as defective. Foreign material includes anything not intended to be present.

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5. PACKAGING.

5.1 Preservation and packaging. Presentation and packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A. Cleaning, drying, preservation, and packaging methods for level A shipment shall be in accordance with instructions contained in applicable documents listed in DL 1330422 (see 6.5.12).

5.1.2 Level C. Cleaning, drying, preservation and packaging shall be in accordance with the contractors commercial practice, when such practices have been approved by the contracting activity (see 6.5.13).

5.2 Packing. Packing shall be level A or C as specified (see 6.2).

5.2.1 Level A. Multiples of the Logic Assembly, preserved and packaged in accordance with 5.1.1, shall be packed in accordance with MIL-P-116, Method III (see 6.5.12).

5.2.2 Level C. The Logic Assembly, packaged as specified in 5.1.2, shall be packed to afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. Containers shall comply with common carrier regulations applicable to the mode of transportation (see 6.5.13).

5.3 Marking.

5.3.1 Normal marking. In addition to markings required by the contract or purchase order, unit packages and shipping containers shall be marked in accordance with DL 1330422.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The Logic Assembly is intended for use in a torpedo exercise head.

6.1.1 Description. The Logic Assembly is a plug-in module that is installed in the instrumentation assembly of the exercise head Mk 85. The electronics are of all solid state design and contain circuitry required for instrumentation and torpedo recovery functions. Specific instrumentation circuits include: course gyro demodulator, time mark generator, - 10 Vdc instrumentation voltage, and RPM tachometer circuitry. Torpedo recovery circuits include torpedo turnaway, inertia switch, lead dropper, squib, and squib firing control circuits.

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6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1, and 2.1.2).

c. Material and treatment if other than as specified (see 3.2).

d. Type of unit (see 1.2 and 3.3).

e. If a first article sample is required (see 3.1).

f. Assigned activity for first article and periodic production test (see 4.5 and 4.6).

g. Environmental test conditions if other than as specified (see 3.5 and 4.3).

h. All additional samples required shall be provided and inspected at the expense of the contractor, when such additional samples are required because of failure of any sample to pass the prescribed inspections (see 4.5, 4.6 and 4.7).

i. If test report required (see 4.4).

j. Number of Logic Assemblies required.

k. Required level of packaging and packing (see 5.1 and 5.2).

l. If sequence of quality conformance testing is not to be as shown in 4.7.

m. Markings required (see 5.3.1).

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Description (DIDs) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DIDs are tailored to reflect the requirements of the specific acquisition, To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DoD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference paragraph	DID Number	DID Title	Suggested Tailoring
4.4	DI-T-2072	Reports, Test	

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The above DIDs were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DIDs are cited on the DD Form 1423.

6.4 First article. When a first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first production items, a standard production item from the contractor's current inventory (see 3.1), and the number of items to be tested as specified in 4.5. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.5 Definitions. For the purpose of this specification, the following definitions have been established.

6.5.1 Nonoperating. The condition of the uninstalled Logic Assembly when stored, transported, or handled (see 5.1.1).

6.5.2 Ready nonoperating. The condition representative of the Logic Assembly when assembled with end item, transported and handled.

6.5.3 Ready, The condition representative of the Logic Assembly when fully assembled with the end item and installed on the launching mechanism prior to operating.

6.5.4 Operating. Operating is the condition of the Logic Assembly when performing its required function.

6.5.5 Vibration. Vibration is a sinusoidal motion defined by amplitude and frequency. The amplitude may refer *to* displacement of acceleration. The displacement is given in inches peak-to-peak. The acceleration is given in multiples of gravitational acceleration (g's). The frequency is given in cycles per second (Hz).

6.5.6 Shock. Shock is an abrupt change in the position of a body, resulting from the sudden application of a large external force of short duration. Shock is described as a pulse in multiples of gravitational acceleration (g's), of duration in units of time, milliseconds (ms), and in the form of a shock pulse approximating a one-half sine wave. The time duration shall be measured between two points where two lines, tangent at maximum slope, intersect the zero g line.

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6.5.7 Acceleration (steady state). Acceleration (steady state) is the time of change of velocity of a body, resulting from the application of a sustained external force with a uniform build-up to peak in more than 0.010 second. Acceleration is given in multiples of gravitational acceleration (g's).

6.5.8 Temperature. The temperature of the environment to which the Logic Assembly may be exposed, expressed in degrees Fahrenheit (°F).

6.5.9 Pressure. The pressure of the environment to which the Logic Assembly may be exposed, expressed in pounds per square inch absolute (lb/in^2).

6.5.10 Humidity. The relative humidity of the environment to which the Logic Assembly is exposed, expressed as a percent.

6.5.11 Defective unit. A unit shall be considered defective if it does not meet all acceptance examinations or tests specified, or deviations are found through visual examination after environmental exposure.

6.5.12 Level A packaging. Maximum protection, designated as level A, is the level of presentation or packing required for protection of material against the most severe worldwide shipment, handling, and storage conditions. Presentation and packing so designated will be designed to protect material against direct exposure to extremes of climate, terrain, operational and transportation environments without protection other than that provided by the pack. The conditions to be considered include, but are not limited to:

a. Multiple handling during transportation and in-transit storage from point of origin to ultimate user.

b. Shock, vibration and static loading during shipment.

c. Loading on shipdeck, transfer at sea, helicopter delivery and offshore or over-the-beach discharge, to ultimate user.

d. Environmental exposure during shipment or during in-transit operations where port and warehouse facilities are limited or nonexistent.

e. Outdoor storage in all climatic conditions for a minimum of 1 year.

f. Static loads imposed by stacking.

6.5.13 Level C packaging. Minimum protection, designated as level C, will be used for protection of material under known favorable conditions. In general, the following criteria determine the requirements for this level of protection:

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a. Use or consumption of the item at the first destination.

b. Shock, vibration and static loading during the limited transportation cycle.

c. Favorable warehouse environment for a maximum of 18 months.

d. Effects of environmental exposure during shipment and in-transit delays.

e. Stacking and supporting superimposed loads during shipment and temporary storage.

When level C is specified, it must reference applicable Public Laws (Code of Federal Regulations) or a specific Federal or Military Specification, Standard or Instruction.

6.5.14 Industrial packaging. Industrial packaging will be acceptable for any level of protection whenever the technical design details of the package meet all conditions of the level of protection specified. Industrial packaging must provide the same level of protection against physical and environmental damage as the military package. It will be marked to the level it meets. Bulk type practices such as are used in inter and intraplant shipments or shipments to jobbers are not acceptable, unless they are usual trade practices for selected commodities.

6.5.15 Storage Life. The Logic Assembly preserved and packaged in accordance with Section 5, is intended to withstand an unattended storage life of 5 years.

6.6 Subject term (key word) listing.

Torpedo
Exercise Head

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing Activity
Navy - OS
(Project 1355-N482)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1 DOCUMENT NUMBER	2 DOCUMENT TITLE
MIL-L-82452A (OS)	LOGIC ASSEMBLY (For Exercise Head MK 85)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)* _____

5 PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording.

c. Reason/Rationale for Recommendation.

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8 DATE OF SUBMISSION *(YYMMDD)*

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